

1960

35,797 FEET

THE DEPTH REACHED BY THE *TRIESTE*



The submersible *Trieste* was designed to withstand the immense pressures of the deepest part of the Marianas Trench. Until 2012, it was the only manned vessel to get there.



LOUIS LEAKEY (1903–72)

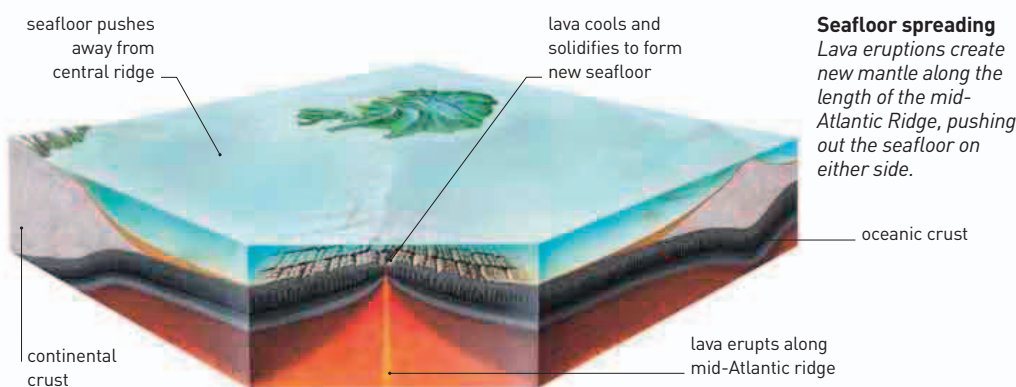
British anthropologist Louis Leakey was instrumental in advancing the understanding of human evolution. Together with his wife, Mary, he spent much of his career studying fossils in East Africa. He showed that humankind originated in Africa, and later in his career helped inspire the work of primatologists such as Jane Goodall and Dian Fossey.

Anthropologists **Louis and Mary Leakey** had been excavating prehistoric stone tools at Olduvai Gorge in East Africa for two decades. In July 1959, Mary found a prehistoric skull. It belonged to an **australopithecine**, an ancestor of modern humans who was later thought to be the first “**apeman**” to use stone tools.

JUST AFTER MIDDAY ON JANUARY 23, the US Navy submersible *Trieste* touched down at the **deepest spot of the world’s oceans: Challenger Deep** in the Marianas Trench of the western Pacific Ocean. The submersible’s onboard team of **Don Walsh** (b.1931) and **Jacques Piccard** (1922–2008) spent 20 minutes there, and saw that some animals were adapted to survive even at these great depths.

Harry Hess (1906–69), a geologist with the US Navy, had studied the ocean depths during World War II. In 1960, he suggested that **the entire seafloor was moving**. He later stated that the molten magma spewing out of Earth’s crust from underwater ridges was cooling, expanding, and pushing the oceanic plates on either side. Today, Hess’s theory is accepted by geologists. It is believed that as new mantle forms at a ridge, old mantle plunges back into Earth elsewhere. This process is responsible for **continental drift**, a theory first suggested by Alfred Wegener (see 1914–15) nearly 50 years before.

In April, NASA launched the **first successful weather satellite, TIROS-1** (Television Infrared Observation Satellite Program-1). For 78 days, television cameras aboard the satellite took thousands



of images of cloud cover and other aspects of atmospheric conditions from aerospace.

In August, American physicist **Theodore Maiman** (1927–2007) demonstrated a new way of producing a concentrated “pencil-beam” of light known as a **laser** (Light Amplification by Stimulated Emission of Radiation, see 1957).



His method involved using a rod of synthetic ruby to produce a series of laser pulses. This technology was later modified to produce a continuous beam that today has applications ranging from eye surgery to compact disc players and supermarket scanners.

American chemist **Robert Woodward** (1917–79) had spent the last decade studying the chemical structures of complex biological substances, such as cholesterol and quinine. He showed that the rules of structural chemistry could be used to produce these substances in the laboratory. In 1960, he

First weather satellite
TIROS-1 carried television cameras and photographed Earth’s weather patterns from a height of at least 435 miles (700 km).

succeeded in **artificially creating chlorophyll II**—one of the main components of the green plant pigment that traps light energy for photosynthesis.

In October, the **11th General Conference on Weights and Measures** published a series of unit standards. Known as **Le Système International d’Unités** (SI units), these were adopted by scientists and technologists.

Also in October, British surgeon **Michael Woodruff** (1911–2001) performed the **UK’s first kidney transplant operation**. It was performed between identical twins to minimize the risk of rejection. Both donor and recipient survived the operation and went on to live many years.

August Chinese-American biologist **Min Chuen Chang** successfully performs **in vitro** fertilization in rabbits

January 23 **Don Walsh** and **Jacques Piccard** descend into the **Marianas Trench**—the deepest spot on Earth

April 1 The US launches its first weather satellite—**TIROS-1**

August 6 American physicist **Theodore Maiman** develops first operational **laser**

October The **SI system** for standard metric measurement established

October 7 The Soviet **Lunar 3** probe photographs the far side of the Moon

July American chemist **Robert Woodward** publishes his work on the **artificial synthesis of chlorophyll**

October 20 British surgeon **Michael Woodruff** performs **UK’s first kidney transplant**

American geologist **Harry Hess** presents his theory of **seafloor spreading** to the US Office of Naval Research